

Technology Stack

Date	05 July 2026
Team ID	
Project Name	Green-Predict-Pro
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, Bootstrap, JavaScript	Responsive, lightweight UI suitable for farmers using low-end mobile devices.
2	Backend / Server-Side	Python 3.x, Flask	Fast development, huge ML ecosystem, easy integration with scikit-learn models.
3	Database / Data Storage	SQLite (development), MySQL (production); CSV crop dataset	SQLite for local prototype; MySQL scales for production; CSV keeps the training data portable.
4	Cloud / Hosting / Deployment	Render / Heroku (Flask app), GitHub for code	Cost-effective free tiers, quick deployment from GitHub.
5	Version Control & CI/CD	Git & GitHub	Seamless team collaboration and version history.
6	Third-Party APIs / Other Tools	NumPy, Pandas, scikit-learn, Matplotlib, Seaborn, SciPy	Provide data processing, model training, evaluation and visualization capabilities.